Assignment 2- Random Forest

Credit default refers to the failure of a borrower to fulfill their financial obligations as agreed upon in a loan or credit agreement. It occurs when the borrower becomes unable or unwilling to make timely payments towards their debts, leading to a breach of contract. Credit defaults can have significant consequences, including negative impacts on the borrower's credit score, financial stability, and the lender's potential financial losses. Lenders often take measures to assess and manage credit default risks, such as credit scoring, collateral requirements, and monitoring borrower's creditworthiness. In this assignment, we want to design a decision tree that can predict if a costumer is suspicious to default the credit or not. You can get more information about the data here.

Now, follow these steps:

0) Assign y to the “class” variable and X to the rest of the variables.

1) Split the data into 80% for training and 20% for test

2) Create a Random Forest model using training data

3) Test the model using Test data, show confusion matrix and measure the accuracy

4) Write the four most important features of this dataset

Attribute Information:

This research employed a binary variable, default payment (Yes = 1, No = 0), as the response variable. This study reviewed the literature and used the following 23 variables as explanatory variables:

- ID: Costumer’s ID
- LIMIT_BAL: Amount of the given credit (NT dollar): it includes both the individual consumer credit and his/her family (supplementary) credit.
- SEX: Gender (1 = male; 2 = female).

- EDUCATION: Education (1 = graduate school; 2 = university; 3 = high school; 4 = others).
- MARRIAGE: Marital status (1 = married; 2 = single; 3 = others).
- AGE: Age (year).
- PAY_0~6: History of past payment. We tracked the past monthly payment records (from April to September, 2005) as follows:
- PAY_0: the repayment status in September, 2005; PAY_2 = the repayment status in August, 2005; . . .; PAY_6 = the repayment status in April, 2005. The measurement scale for the repayment status is: -1 = pay duly; 1 = payment delay for one month; 2 = payment delay for two months; . . .; 8 = payment delay for eight months; 9 = payment delay for nine months and above.
- BILL_AMT1~6: Amount of bill statement (NT dollar). BILL_AMT1 = amount of bill statement in September, 2005; BILL_AMT2= amount of bill statement in August, 2005; . . .; BILL_AMT6 = amount of bill statement in April, 2005.
- PAY_AMT1~6: Amount of previous payment (NT dollar). PAY_AMT1 = amount paid in September, 2005; PAY_AMT2= amount paid in August, 2005; . . .; PAY_AMT6 = amount paid in April, 2005.
- DEFAULT: Default the payment of next month

You are free to use either R or Python for this assignment, but the submission should be in HTML format. Note that if the variable is in the wrong data type. Make sure to convert the target variable into the appropriate type.

This assignment contains two files:

1) Instructions (this file)

2) credit.csv